

Foundation (Jalapeno)

Q1. Simplify $2 \cdot \sqrt{8}$

- (A) $4\sqrt{2}$
- (B) $4\sqrt{3}$
- (C) $2\sqrt{4}$
- (D) $8\sqrt{2}$
- (E) $8\sqrt{3}$

Q2. Add $3\sqrt{5}$ and $2\sqrt{5}$

- (A) $5\sqrt{3}$
- (B) $5\sqrt{5}$
- (C) $3\sqrt{3}$
- (D) $6\sqrt{2}$
- (E) $10\sqrt{2}$

Q3. Subtract $5\sqrt{3}$ from $8\sqrt{3}$

- (A) $3\sqrt{2}$
- (B) $3\sqrt{3}$
- (C) $12\sqrt{2}$
- (D) $13\sqrt{3}$
- (E) $10\sqrt{3}$

Q4. Simplify $(2\sqrt{6})(3\sqrt{6})$

- (A) $30\sqrt{2}$
- (B) $36\sqrt{6}$
- (C) $36\sqrt{2}$
- (D) $12\sqrt{3}$
- (E) $6\sqrt{12}$

Q5. A stock's price is $12\sqrt{2}$. If it increases by $4\sqrt{2}$, what is the

new price?

- (A) $8\sqrt{2}$
- (B) $16\sqrt{2}$
- (C) $12\sqrt{2}$
- (D) $10\sqrt{2}$
- (E) $20\sqrt{2}$

Q6. Rationalize the denominator of $\frac{2}{\sqrt{5}}$

- (A) $\frac{2\sqrt{5}}{5}$
- (B) $\frac{5\sqrt{2}}{2}$
- (C) $\frac{2\sqrt{2}}{5}$
- (D) $\frac{10}{\sqrt{5}}$
- (E) $\frac{5}{2\sqrt{5}}$

Q7. Which of the following is equivalent to $\sqrt{18}$?

- (A) $3\sqrt{2}$
- (B) $3\sqrt{3}$
- (C) $6\sqrt{2}$
- (D) $9\sqrt{2}$
- (E) $9\sqrt{3}$

Q8. A company's tax is $250\sqrt{2}$. If it decreases by $50\sqrt{2}$, what is the new tax?

- (A) $150\sqrt{3}$
- (B) $200\sqrt{2}$
- (C) $250\sqrt{3}$
- (D) $250\sqrt{2}$
- (E) $300\sqrt{2}$

Open Ended Questions (FRQ)

Q9. Simplify $(6\sqrt{3})(2\sqrt{3})$ and show your work.

Q10. A savings account earns $200\sqrt{2}$ in interest per year. If the interest rate increases by $50\sqrt{2}$, what is the new interest earned and show your calculation?

Chef's Tips

- Tip 1: Use a calculator to check answers only when necessary
- Tip 2: Emphasize the importance of rationalizing denominators
- Tip 3: Encourage students to show all work for free response questions